

Novelty and Prior-Art Audit

Standalone preliminary review | David Bridges research | 1 October 2026

Conclusion

No direct prior match was identified in the sources inspected for the exact normalized principal-upset counterexample or its complete two-realization fiber classification. Their novelty remains unresolved. The search does not establish priority, absence from the literature, minimality, publishability or expert acceptance.

The two candidate contributions

Candidate A: explicit nonisomorphic finite posets P, Q with identical full normalized principal-upset Gram matrices G . The retained example has 16 elements in each poset. Candidate B: complete classification of that particular labeled G , with exactly P and Q as realizations, distinguished by their comparability-degree multisets. Correctness is anchored to the frozen v127/v129 proof and reproducibility audits; this document is a literature comparison, not a new mathematical proof audit.

What is already familiar

Zeta and incidence matrices, Boolean meet/join identities, diagonal matrix factorizations, normalized vector inner products, and nonuniqueness of general Gram factors are established ingredients. The elementary saturation, degree-sum and singleton-maximum formulas should be presented as derivations or supporting lemmas, rather than independent discoveries on the evidence of this search.

A careful novelty assessment

The potentially distinctive content is realizing the normalized ambiguity inside the restrictive class of full finite-poset zeta matrices, proving nonisomorphism, and exhaustively identifying this particular fiber. It is not the word Gram, the use of Boolean sets, or the existence of arbitrary matrices with equal Gram products. The construction must be compared with known mechanisms, not only matched by terminology.

Status

Potentially original explicit results: not certified. Established or elementary ingredients: acknowledge. General G -plus-degree injectivity: unresolved in the project. Physical discovery: not supported. This standalone document changes no canonical claim, packet or repository.

1. Claim-by-claim assessment

Novelty is separate from mathematical correctness.

A. Exact 16-element normalized collision

No exact antecedent was located in inspected material. The invariant is $G=D^{(-1/2)}ZZ^TD^{(-1/2)}$, where Z is the full reflexive order matrix and D contains principal-upset sizes. Equality is exact and P, Q are nonisomorphic. Generic incidence trades are relevant precedent but are not automatically the same result. Verdict: candidate contribution, priority unresolved.

B. Full classification of this particular fiber

All 16 possible maximum anchors were checked, with exactly two valid realizations. This is a concrete complete computation supported by a completeness proof. No matching prior classification was located. The method is elementary and the specific example may be new or a routine consequence of older work; this review cannot distinguish those possibilities conclusively. Verdict: candidate companion result, novelty unresolved.

C. Positive-entry maximum-anchor method

Positive G excludes distinct maxima, and $d_i=1/G_{im}^2$ at a proposed maximum. Saturation and closure then exhaust candidates. These formulas follow directly from principal upsets. No dedicated antecedent was identified, but that does not justify calling the method a new theorem in the literature. Verdict: supporting enumeration lemma; do not assert priority.

D. Degree filter and scale-sum identity

The formula $\text{degree}(i)=\text{upset_size}(i)+\text{downset_size}(i)-2$ and the sum identity are direct incidence counting. Adding the computed degree multiset to candidate closure is a natural filter. Verdict: supporting identities, no independent novelty claim.

E. Dirac comparisons and finite generator tests

Hodge-Dirac operators are established constructions. The particular spectral distinction is an exact project computation, but its priority was not separately audited here. B3/B4 generator tests are finite evidence, not universal reconstruction proofs. Neither category is used to certify A or B as novel. The physical Dirac equation is not derived.

2. Closest inspected precedents

Comparison by invariant and mechanism.

[1] Ballantine, Frechette and Little

Their paper studies $Z+Z^T$ and its determinant, including Boolean cases; sections 1 and 4 were consulted. This differs from the normalized product $D^{(-1/2)}ZZ^TD^{(-1/2)}$. Its existence prevents any claim that poset zeta-matrix analysis or Boolean determinant work is new. No direct normalized reconstruction antecedent was identified in those inspected portions.

[2] Ilmonen and Kaarnioja

The meet/join paper surveys incidence factorizations and gives a diagonal relation between join and meet matrices under semimultiplicativity; section 2.3 was inspected. This is an important mechanism precedent: diagonal transport and meet/join duality are familiar. It does not, in the inspected portions, provide the full normalized two-sector poset collision or its fiber classification. A deeper comparison with cited original meet/join sources remains necessary before claiming structural originality.

[3] Gasiorek

The abstract explicitly uses the Gram terminology for $(C+C^T)/2$, with C encoding order, and classifies under integer congruence. That invariant and equivalence relation differ from our normalized full incidence product and poset isomorphism. Only the abstract was used in this audit; no whole-paper absence claim is made.

[4] Dai and collaborators, differential-poset preprint

Section 2 preserves a rank-cover row Gram matrix through a crown trade: a triple and three singleton blocks are replaced by three pair blocks. The paper attributes that mechanism to earlier work by Wagner and Lewis. This establishes related incidence-Gram ambiguity, so a generic claim to discovering Gram-preserving trades would be inappropriate. Their rank-cover matrix is different from the full normalized principal-upset matrix; the inspected argument does not itself establish our exact claim. This is a preprint, not validation of every asserted result.

Comparison limit

Different matrices do not prove independence of mechanisms. An older construction might imply ours after a transformation or lifting. That implication was not found, but has not been exhaustively excluded.

3. Search coverage and gaps

Bounded public-web audit, not a literature-wide certification.

Coverage

Sixteen new queries were executed in four batches across two search systems. They cover normalized Gram and zeta terminology, principal upsets/filters, cosine and correlation language, intersection matrices, maximal anchors, and the cited zeta-matrix authors. The prior v128 30-query working review was consulted as background. These totals describe search activity, not independent relevant publications.

What was actually inspected

Primary passages were retrieved from [1], [2] and [4]; [3] was inspected at abstract level. A publisher/library page for Cover matrices of posets and their spectra was opened, but its full paper was not assessed, so it is not used to exclude an antecedent. Unrelated results were discarded rather than counted as corroboration.

Unresolved leads

A search result advertised an unnormalized divisor-poset counting Gram/precision construction under the title Pythagorean Angle Lattice: Mobius Precision, Spectral Addition and Depth Reconstruction. Only a third-party snippet was seen; its underlying paper was not authenticated or reviewed. It is a lead, not evidence for or against novelty. The original meet/join references cited in [2], and the older crown-reflection sources cited in [4], require further direct inspection.

Database and retrieval gaps

No systematic MathSciNet or zbMATH review, complete Google Scholar citation traversal, all-language search, thesis archive search, or exhaustive bibliographic backward/forward scan was completed. Search indexes may omit relevant phrases, papers or equivalent formulations. A failed find for Summary on the cover-matrix landing page is retained in the raw retrieval record and is not evidence that the paper lacks a relevant result.

Record fidelity

query_log.json lists all sixteen literal queries. Actual returned JSON for calls 2-5 is retained in the supporting record. The first returned response remains in the chat tool record and was not reconstructed as an exact saved response. No private expert opinion, peer review or correspondence response is claimed.

4. Permitted wording and next audit step

A conservative conclusion that can be shared.

Suggested description

We give an explicit 16-element pair of nonisomorphic posets with identical normalized principal-upset Gram matrices and classify all realizations of the resulting matrix. A bounded literature review has not identified a direct antecedent; novelty and priority remain unresolved.

Claims this audit does not authorize

Do not write first ever, previously unknown, smallest counterexample, new physical principle, or novel universal reconstruction theorem. Do not imply that the richer G-plus-degree question is open throughout the literature merely because our project has not answered it. Expert review and broader bibliographic checking could change the novelty assessment without changing the verified counterexample.

Most useful next literature work

First inspect the original meet/join diagonal-duality references cited in [2] and the crown-reflection sources cited in [4]. Determine whether their mechanisms yield a full normalized-zeta collision after a known construction. Then seek subject-expert references or a database-assisted citation audit. No outreach was sent as part of this task.

References

- [1] C. M. Ballantine, S. M. Frechette, J. B. Little, Determinants Associated to Zeta Matrices of Posets (2004 preprint; Linear Algebra Appl. 411, 2005, 364-370). <https://arxiv.org/html/math/0403401v1>
- [2] P. Ilmonen, V. Kaarnioja, Generalized eigenvalue problems for meet and join matrices on semilattices (2017 revision). <https://arxiv.org/html/1705.05169v2>
- [3] M. Gasiorrek, Structure of non-negative posets of Dynkin type A_n (2023 revision). <https://arxiv.org/abs/2205.15032>
- [4] A Negative Answer to Stanley's Problem 4 on Differential Posets (2026 preprint). <https://arxiv.org/html/2607.24541v1>

Internal source

Frozen Audit 10 v129 and the reader edition establish the mathematical claims summarized here. This review is a separate standalone document. No novelty certification is added to those checkpoints.